Genetic Plasticity in Extreme Environments: Comparative Genomic Analysis of Extremophiles Using *aurora*

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ABSTRACT:

Extremophiles offer insight into how organisms adapt under strong selective pressure, but it remains unclear whether distantly related extremophile lineages undergo convergent genomic change in similar functional pathways. This study applied *aurora*, a machine learning-based microbial genome-wide association tool, to two independent extremophile cohorts, thermophilic and mesophilic Bacillaceae (n=32), and acidophilic and non-acidophilic Alicyclobacillaceae (n=41), to identify habitat-associated genomic regions while controlling for phylogenetic confounding. Genomes were annotated with Prokka, compiled into pangenomes with Panaroo, and analyzed for gene-habitat association using random forest and random walk models. No gene family reached Bonferroni-corrected significance in either cohort. However, the top-ranked thermophile genes, *ctsR* and *mcsB*, are known regulators of heat-shock stress response.These results suggest that habitat adaptation signal may be biologically real but diffuse, polygenic, or limited by genome assembly quality and taxonomic breadth rather than completely absent. Findings are discussed in the context of *aurora*'s prior validation on colonization traits, extremophile genomic resource limitations, and implications for astrobiological models of extreme-environment adaptation.

KEYWORDS: *aurora,* extremophiles, comparative genomics, machine learning

INTRODUCTION:

Both novel innovations and gradual adaptations of traits and pathways have allowed a diverse range of extremophiles to develop within their niche environments (Chasapi et al., 2026). It is worth considering whether there are common pathways that are likely to undergo change in extreme environments, even if they aren't adapting in the same ways. Are extremophiles of different environments experiencing evolutionary change within similar regions of the genome? Given that certain cellular functions are fundamental to life processes, it is anticipated that genome regions involved in basic cellular processes have been adapted across most extremophiles, though the changes are unlikely to be identical.

Extremophiles are a broad group of organisms that thrive in intense environments that are too inhospitable for most life, such as hot hydrothermal vents, sub-glacial lakes, and soda springs. Prior research has already found various mutations in extremophiles that encode for proteins and enzymes that are able to function in extreme environments, called extremozymes (Chasapi et al., 2026). For example, thermophiles have proteins that are stabilized by an increase in hydrophobic structures, more disulfide bonds, and ionic interactions, which result in proteins that have a higher melting point and do not as easily undergo denaturation (Brininger et al., 2018). These protein changes are able to be mapped to distinct genetic mutations.

Organisms that live in these extreme environments survive because their proteins have evolved different amino acid compositions resulting in structural, and

thereby functional, features that allow them to perform in the environment. The adjustments required to overcome these challenges vary based on the environment. Malate dehydrogenase (MDH) is a well-characterized protein and has been used to observe environmentally induced adaptations that allow proteins to function (Brininger et al., 2018). Additionally, Brininger et al. (2018) found that more general extremophile protein adaptations are conserved and can be predicted in extremophilic proteins. This is promising and shows that there are general evolutionary patterns across different extremophile groups

However, this project shifts the focus from individual species adaptation to a broader pathway convergent perspective among extremophiles. A comparative genomic analysis among species of diverse environments allows for greater understanding of evolutionary changes, especially under strong selective pressures found at the extremes of what life can withstand. Additionally, this research could provide insight into what genetic plasticity is still present even among highly conserved regions of the genome responsible for fundamental cellular processes.

While extremophiles offer unique insight into evolutionary adaptation, they are difficult to study. Laboratory limitations don't easily allow for extremophile study in conditions to which they are adapted, which makes culturing, sampling (especially while preserving the life of sampled cells), and biomolecule extraction difficult (Chasapi et al., 2026). For these reasons, current research relies on metagenome-assembled genomes (MAGs), where an environmental sample is taken and all DNA in the sample is sequenced; an algorithm then bins those sequences into individual genomes (Chasapi et al., 2026). While this is a useful workaround, there is a limit of how much information can be gathered from MAGs. Due to a gap in extremophile genome annotation, associating novel sequences with functional proteomic data is difficult. The lack of functional mapping of extremophile genomes means it is not fully understood how broader genetic and cellular pathways are adapting across different lineages.

Beyond MAGs, whole genome shotgun sequencing (WGS) has been used to analyze entire microbial genomes. This metagenomic approach mitigates the need for microbes to be cultured in the lab in order to be sequenced and analyzed. Additionally, WGS allows for community level analysis rather than a single organism and also has the potential to discover new genes that are too diverged from currently known genes to be amplified with PCR (Chen and Pachter, 2005). However, Chen and Pachter (2005) found that there was difficulty comparing multiple distinct lineages because highly conserved genes are heavily shared across species, resulting in sequence overlap and potential algorithmic confusion. Sharing a similar difficulty, microbial Genome-Wide Association Studies (mGWAS) aim to find a statistical association between gene variants and specific phenotypes. Highly overlapping sequences are a confounding variable, and mGWAS algorithms can return high false positive rates. Additionally, microbial genomes undergo rapid evolution and display high plasticity, which with horizontal gene transfer and asexual reproduction leads to a fluctuating genome and genome-wide

linkage disequilibrium (Bujdoš et al., 2025). It is difficult to infer statistically significant associations in microbial genomes for these reasons.

Unlike other mGWAS, the machine learning-based tool *aurora* (AUtochthonous, Random fOrest, Random walk) is able to identify genes associated with certain habitats even if the variants shaped the phylogeny being analyzed (Bujdoš et al., 2025). It seems that this tool is not widely used in extremophile research. This realm would especially demonstrate *aurora*'s usefulness, as it is a prime example of lineages that are shaped by gene variation associated with habitat. Therefore, the purpose of this study is to apply *aurora* to multiple extremophile groups to map common pathway adaptation locations. The study seeks to understand what metabolic and cellular pathways display evolutionary plasticity, regardless of whether the evolutionary changes are the same across extremophile groups. It is hypothesized that *aurora* will be able to identify distinct, nonrandom locations of genetic variation associated with habitat specialization within conserved cellular pathways across divergent lineages.

METHODOLOGY:

This quantitative approach utilized bacterial genome sequence data sourced from NCBI RefSeq to ensure consistent, cross-compatible sequence formats across different species. Initial target identification was supplemented by pre-existing extremophile databases, including AciDB (Neira et al., 2020), ThermoBase (DiGiacomo et al 2022), which aggregate comprehensive data on acidophiles and thermophiles, respectively.

To optimize the machine learning performance of the downstream random forest models, a target cohort framework was established requiring a minimum of 30 total genomes per target extremophile/mesophile group, ideally maintaining a 1:1-1:3 ratio. To reduce genetic noise and control for confounding variables during pangenome alignment, selection for each independent cohort was strictly limited to species within the same taxonomic family.

To evaluate pipeline feasibility, an exploratory data analysis (EDA) was performed, targeting thermophilic bacteria within the family *Bacillaceae*. A Python script, ExtremophileCounter, analyzed the species in ThermoBase, confirming the presence of 93 thermophile Bacillaceae species. Then a second script, GenomeSelector, was run. This was developed to cross-reference this list against 3,323 complete RefSeq genomes returned by the NCBI search. GenomeSelector isolated taxonomic elements into distinct "Genus" and "Species" variables to account for spelling, strain, and naming variations between databases. After removing duplicate genomes of the same species, 8 complete thermophile genomes were isolated (Table 1).

For each thermophile genome, GenomeSelector randomly selected three mesophilic species within the same genus to form the control group. In instances where less than three non-extremophile species of the same genus were present, any that were

available were selected. When no matching mesophiles for a given thermophile genus were present, three random but unique *Bacillaceae* mesophiles were chosen, yielding a final control group of 24 mesophilic species (Table 2). Empirical limits identified during this EDA indicated that even strongly researched extremophile groups may yield fewer complete genomes than originally anticipated. Consequently, project expectations were adjusted to focus strictly on the most heavily documented extremophile groups, including thermophiles, and acidophiles. The previous steps were all repeated for the acidophile group, including a secondary EDA for quality assurance.

Upon selection of cohort species and controls, various processing steps took place. All target sequences were retrieved in FASTA format. Because downstream pangenome creation relies on already existing functional annotations, raw FASTA sequences were compiled and annotated using Prokka, a command-line software tool for prokaryotic genome annotation, yielding standard .gff files (Seemann, 2014). The resulting .gff outputs were processed through Panaroo, a Python-based clustering pipeline that creates pangenomes. Panaroo collapses genes present in the entire cohort into a matrix that displays whether the gene is present or absent in a given species, and shows genome alignments (Tonkin-Hill et al., 2020).

Habitat-associated gene mapping is driven by the R library *aurora*, a machine learning-based microbial Genome-Wide Association Study (mGWAS) tool (Bujdoš et al., 2025). Unlike traditional mGWAS frameworks that generate high false-positives due to linkage disequilibrium and lineage based overlap, *aurora* effectively detangles habitat genomic features from confounding phylogenetic structures. The pipeline executes two primary functions: aurora_pheno() and aurora_GWAS(). Using the phylogenetic tree and pangenome matrix as concurrent inputs, the random forest and random walk models clear allochthonous background genes and calculate specific habitat-association scores. Statistical significance is evaluated via F_1 scores (balancing gene frequency and recall) and standardized residuals (measuring the magnitude of habitat association) rather than traditional p-values.

Genes exhibiting the highest F_1 scores and standardized residuals were isolated for biological interpretation. Because initial annotations provide gene names without a functional categorization, the prokka gene annotation keywords and categories were used to associate the highest F_1 scoring genes with functional significance. Finally, an Over-Representation Analysis (ORA) was conducted to mathematically determine whether genes associated with specific pathways appear more frequently than expected by random chance. Cross-cohort variations were visualized using a heatmap to isolate overlapping regions of evolutionary plasticity across independent environmental extremes.

RESULTS:

Genome Dataset Characteristics

Thermophilic and mesophilic *Bacillaceae* genomes were selected using ThermoBase and NCBI RefSeq, yielding an initial set of 9 thermophile and 24 mesophile genomes. One thermophile accession (GCF_040959945.1, *Metabacillus halosaccharovorans*) was subsequently excluded following identification as a halophile instead of a thermophile, resulting in a final dataset of 8 thermophile and 24 mesophile genomes (n=32).

Metadata for the final thermophile and mesophile cohorts are summarized in Table 1. Extremophile genomes had a mean size of 3.67 Mb (median 3.70 Mb) and mean gene count of 3751.4 (median 3704). Mesophile genomes had a mean size of 4.49 Mb (median 4.33 Mb) and mean gene count of 4519.4 (median 4354.5). Contig N50 values closely matched genome size values across both groups, consistent with complete, non-fragmented assemblies. A scatter plot of genome size against gene count (Figure 1) shows a linear relationship between the two variables across both cohorts and clustering among thermophile data points, though there is overlap among mesophilic points.

Acidophilic and mesophilic *Alicyclobacillaceae* genomes were selected from NCBI RefSeq, yielding a compiled dataset of 41 genomes (19 Acidophile, 22 Non-Acidophile). Unlike the thermophile cohort, selected genomes spanned varying assembly levels, including contig, scaffold, and complete genome assemblies.

Genomic metadata for the acidophile and non-acidophile cohorts are summarized in Table 2. Acidophile genomes had a mean size of 3.37 Mb (median 3.31 Mb) and mean gene count of 3367.0 (median 3267), compared to a mean size of 3.27 Mb (median 3.10 Mb) and mean gene count of 3276.0 (median 3172.5) for non-acidophile genomes. A scatter plot of genome size against gene count (Figure 2) shows a linear relationship, though without distinct clustering by assignment. Assembly level distribution across both cohorts is shown in Table 3.

Pangenome Construction

Pangenome analysis was performed using Panaroo on 32 thermophile/mesophile genomes and 41 acidophile/non-acidophile genomes. A total of 66,123 genes were identified across the thermophile pangenome. None of the genes were classified as core (99–100% of genomes) or soft core (95–99%). 6,619 were categorized as shell genes (15–95%), and 59,504 as cloud genes (0–15%) (Table 4). Due to the absence of genes meeting the default core gene threshold, the core genome threshold was adjusted to 80% genome presence, yielding a core genome alignment of 50 genes across 28 genomes with a total alignment length of 38,455 bp.

The acidophile pangenome comprised 37,372 total genes, of which 0 were classified as core or soft core, 2,454 as shell genes, and 34,918 as cloud genes (Table 4). Applying the same adjusted 80% core genome threshold produced a core genome alignment of 28 genes across 36 genomes with a total alignment length of 21,832 bp.

Phylogenetic Reconstruction

Phylogenies were created using FastTree. The thermophile phylogeny comprised 28 tips with a total branch length of 2.024 and a final log-likelihood of −360,026.064. SH-like support values ranged from 0.304 to 1.000 (n=26 internal nodes), with 25 of 26 nodes showing support ≥0.950.

The acidophile phylogeny comprised 36 tips with a total branch length of 2.090 and a final log-likelihood of −217,897.967. SH-like support values ranged from 0.140 to 1.000 (n=34 internal nodes), with 33 of 34 nodes showing support ≥0.969.

aurora Gene-Habitat Association Analysis and Functional Assignment

Gene-habitat associations were assessed using *aurora*, consisting of random forest and CART classification models with phylogenetic correction. For the thermophile cohort, initial abundance filtering removed 51,598 of 66,123 variants, and ancestral reconstruction filtering removed an additional 909 variants prior to model fitting. Gene-habitat association analysis was performed on 44,040 gene families with valid association statistics. No gene family reached the Bonferroni-corrected significance threshold ($\alpha = 3.67 \times 10^{-6}$). Genes were ranked by F1 score. The top 10 candidate genes are shown in Table 5, with *ctsR* and *mcsB* showing the highest F1 scores (F1 = 0.759, standardized residual = 1.35), both present in 100% of extremophile strains and 60% of mesophile strains. Clustering based on random forest classification is shown as a heatmap in Figure 3.

Functional category analysis and over-representation analysis (hypergeometric test) of the top 10 candidate genes using Prokka-derived annotations identified uncorrected overrepresentation of the Translation/Ribosomal category (2 of 10 genes; 1,331 of 66,123 total genes; p = 0.016) and the DNA/RNA Processing category (2 of 10 genes; 1,775 of 66,123 total genes; p = 0.028) relative to the genome-wide background. Neither category remained significant after Bonferroni correction for six categories tested ($\alpha = 0.0083$). No other functional category showed uncorrected significance.

For the acidophile cohort, initial abundance filtering removed 15,162 of 37,372 variants, and ancestral reconstruction filtering removed an additional 7,725 variants prior to model fitting. Gene-habitat association analysis was performed on 10,845 gene families with valid association statistics. Again, none of the gene families met the Bonferroni significance threshold ($\alpha = 3.45 \times 10^{-6}$). The top 10 candidate genes with the highest F1 scores are shown in Table 6, including *salA*, *abrB*, and several tRNA synthetase and hypothetical protein-coding genes. A heatmap for the acidophile cohort can be found in Figure 4.

In the functional category analysis of the top 10 genes, uncorrected overrepresentation of the Translation/Ribosomal category was found (2 of 10 genes; 1,112 of 37,372 total genes; p = 0.034) relative to the genome-wide background. As with the thermophile results, however, this category is not statistically significant after Bonferroni correction.

DISCUSSION:

This project aimed to determine whether extremophile lineages from different environmental extremes show convergent signs of genomic habitat adaptation within conserved cellular pathways. It was hypothesized that aurora would identify distinct, nonrandom locations of genetic variation associated with habitat specialization across divergent lineages. This hypothesis was not supported at the level of individual gene significance: no gene family reached the Bonferroni-corrected threshold in either the thermophile or acidophile cohort. However, this non-significance does not necessarily indicate an absence of habitat-associated genomic regions. Rather, several findings suggest that the signal may be real but diffuse, underpowered, or both. The core genomes proved difficult to resolve computationally. No genes met the default core or soft-core thresholds in either cohort, and the threshold had to be relaxed to 80% genome presence before any core alignment could be constructed at all (Table 4). This result previews the power and resolution limitations later discussed in this section.

Despite the lack of statistical significance, the two highest-ranked genes in the thermophile cohort, ctsR and mcsB, are functionally well-characterized and directly relevant to thermal stress adaptation. Though functional assignment of ctsR was uncharacterized by the keyword-based classification of Prokka annotation, ctsR is a known transcriptional regulator of the clpC heat-shock operon (Elsholz et al., 2010). Additionally, mcsB encodes a protein arginine kinase that works with the ClpC/ClpP complex to tag misfolded proteins for degradation under heat stress (Elsholz et al., 2010). The coarseness of the gene's functional assignments is demonstrated in the classification of mcsB as well. However, the presence of ctsR and mcsB in 100% of *Bacillaceae* thermophile strains and only 60% of *Bacillaceae* mesophile strains, combined with the highest F1 scores and standardized residuals in the dataset, suggests that aurora identified biologically plausible candidates even though it did not clear a conservative genomewide correction. This is consistent with Brininger et al. (2018), who found that proteinlevel adaptations in extremophiles are often general and predictable across taxa rather than distinct within single lineages. The result of a well-established stress-response gene at the top of the ranking, even without reaching significance, supports the interpretation that real biological signal is present, but that the statistical power of this study was insufficient to formally confirm it.

In both the thermophile and acidophile cohorts, independently, the

Translation/Ribosomal category was overrepresented (uncorrected) among the top ten candidate genes. The recurrence of this category across two independent extremeenvironment cohorts is suggestive of convergent selection acting on core cellular machinery, though when corrected this relationship is not statistically significant. Functional categorization relied on keyword-based classification of Prokka-derived gene annotations (Seemann, 2014) rather than a dedicated orthology and functional-mapping tool such as eggNOG-mapper, which was unavailable during analysis. Prokka annotations provide a reasonable basis for approximate functional grouping, but

keyword-based classification is more prone to misclassification than orthology-based methods. To determine if there is any significance in the representation of

Translation/Ribosomal related genes more accurate functional annotation would be an appropriate next step.

The absence of any Bonferroni-significant gene family in this project is best explained primarily by limited sample size and statistical power, though the underlying causes differed between cohorts. *aurora* was designed to detect habitat-associated genomic signal from confounding phylogenetic structure, even in cases where the trait itself shaped phylogeny (Bujdoš et al., 2025). It is notable, therefore, that even this tool did not return a significant hit in either extremophile cohort. For the acidophile cohort, genomes spanned multiple assembly levels (contig, scaffold, and complete) as shown in Table 3, and a substantially larger proportion of variants were removed during ancestral reconstruction filtering (7,725 of 37,372; 20.7%) compared to the thermophile cohort (909 of 66,123; 1.4%). This disparity suggests that incomplete or fragmented assemblies introduced meaningful noise into the pangenome matrix and phylogenetic inputs, reducing the reliability of the *aurora* results. The thermophile cohort, by contrast, consisted entirely of complete genome assemblies, yet also yielded no significant hits and no genes meeting the default core genome threshold. Here, the explanation is more likely taxonomic: the eight thermophile genomes available for analysis spanned six distinct genera within *Bacillaceae*, a breadth of diversity that may be sufficient for Panaroo's ortholog-clustering approach to fail to resolve shared core genes across the full cohort. This is consistent with the difficulty Chen and Pachter (2005) describe in resolving lineage-level relationships when sequence divergence and taxonomic breadth complicate cross-genome comparison. In other words, the thermophile and acidophile cohorts appear to have been limited in two ways: assembly-quality and taxonomic breadth. Ultimately both cohorts are constrained by the limited number of complete extremophile genomes available for analysis in either group. Unfortunately to compensate for a lack of complete extremophile genomes within close taxonomic proximity, it is necessary to either use fewer genomes, or to accept genomes that are fragmented.

A further open question raised by these results is whether the failure to reach significance reflects insufficient statistical power alone, or if it is also indicative of polygenic mechanisms contributing to habitat adaptation in these lineages. Many genes of modest individual effects could contribute to adaptation rather than a small number of genes with strong effects. This is especially possible in the thermophile cohort, where the top 10 genes display F1 scores ranging from 0.759 to 0.667 (Table 5). Among the acidophile cohort, however, it seems the results are more likely to be due to overfitting. The top 10 genes from *aurora* analysis all have F1 scores of 1.0 (Table 6). The uniformity of these results suggests that the analysis couldn’t meaningfully differentiate between the top gene results. This ceiling effect is consistent with the same assemblyquality issues discussed above. Heavy ancestral reconstruction due to the

assembly level variation and fragmentation removed a substantial number of variants. The left behind group of variants could have been small enough to display artificially clean gene presence data rather than resulting in genes that reflect habitat associations. The difference between the cohorts in this regard is likely explained by the differences in assembly levels.

These findings show broader relevance to astrobiology beyond the immediate microbiological context. Extremophiles are frequently used as analogs for life that might exist in extraterrestrial environments, and bioinformatic and machine-learning approaches have become central to interpreting extremophile genomic data at scale (Chasapi et al., 2026). If habitat adaptation is polygenic rather than concentrated in a few genes, a clean, universal genomic signature of extreme-environment adaptation may be difficult to find at all. This is relevant both to extremophile research on Earth and to the speculation of evolutionary mechanisms elsewhere. This should be interpreted cautiously given the sample sizes involved, but it is worth noting that *aurora*'s published validation was conducted primarily on host-associated colonization traits, such as gut niche adaptation in species including *Limosilactobacillus reuteri*, *Lactiplantibacillus plantarum*, *Salmonella Typhimurium*, and *Mycobacterium avium subsp. paratuberculosis* (Bujdoš et al., 2025), rather than on traits shaped by sustained abiotic stress. Colonization adaptation may plausibly involve a smaller number of genes that have more effect, whereas adaptation to sustained thermal or pH extremes across deep phylogenetic divergence may involve a fundamentally different, more distributed genetic mechanisms for adaptation. If so, the null result reported here may reflect a mismatch between the genetic variation *aurora* was built and validated to detect and the genetic variation present in extremophile abiotic stress adaptation. This possibility is further supported by Brininger et al. (2018), whose finding that extremophile protein-level adaptations are general and conserved across taxa is consistent with a distributed, rather than concentrated, model of adaptation.

Several directions for future research follow directly from these findings. First, *aurora* could be explicitly benchmarked against genes with well-established roles in abiotic stress adaptation, such as ctsR and mcsB, to determine whether *aurora* struggles to detect polygenic trait architecture, regardless of the sample size. Second, expanding cohort sizes as more complete extremophile genomes become available, especially among more taxonomically close species, would help resolve the taxonomic-breadth and statistical significance issue observed in the thermophile cohort. Third, re-running functional categorization using a dedicated orthology tool such as eggNOG-mapper or Gene Ontology-based annotation (The Gene Ontology Consortium et al., 2023), rather than keyword-based Prokka classification, would allow the functional categorization patterns to be analyzed with greater confidence.

CONCLUSION:

This study set out to determine whether extremophile lineages from different environmental extremes show convergent genomic signatures of habitat adaptation within conserved cellular pathways. While *aurora* did not identify any gene family reaching Bonferroni-corrected significance in either the thermophile or acidophile cohort, the presence of well-characterized stress-response genes (*ctsR*, *mcsB*) at the top of the thermophile ranking, along with the independent overrepresentation of Translation/Ribosomal genes in both cohorts, suggests that real biological signal may be present but diffuse, polygenic, or obscured by limited genome availability and assembly quality. These findings highlight the challenges of detecting habitat adaptation signal in extremophile genomes using existing mGWAS tools, and point to concrete next steps: benchmarking *aurora* against known abiotic stress genes, expanding taxonomically close cohort sizes, and applying orthology-based functional annotation. More broadly, this work underscores that genomic adaptation to environmental extremes may not resolve into simple, universal signatures. This consideration is relevant to both extremophile biology and astrobiological models of life in extreme conditions.

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TABLES AND FIGURES:

Table 1:

Summary of Selected Bacillaceae Species' Genomic Metadata

Target Cohort	Total Genomes	Distinct Genera	Mean Size (Mb)	Median Size (Mb)	Mean Gene Count	Median Gene Count	Mean N50 (Mb)	Median N50 (Mb)
Extremophile	8	6	3.67	3.70	3751.375	3704	3.67	3.69
Mesophile	24	6	4.79	4.53	4833.542	4601	4.69	4.51

Table 2

Summary of Selected Alicyclobacillaceae Species' Genomic Metadata

Target Cohort	Total Genomes	Mean Size (Mb)	Median Size (Mb)	Mean Gene Count	Median Gene Count	Mean N50 (Mb)	Median N50 (Mb)
Acidophile	19	3.37	3.31	3366.895	3267.0	0.62	0.13
Non-Acidophile	22	3.27	3.10	3276.045	3172.5	0.94	0.15

Table 3:

Assembly Level Distribution of Selected Alicyclobacillaceae Genomes

Assembly Level	Acidophile	Non-Acidophile
Complete Genome	3	5
Contig	9	15
Scaffold	7	2

Table 4:

Pangenome Gene Category Distribution by Cohort		
Gene Category	**Thermophile Cohort**	**Acidophile Cohort**
Core (99–100%)	0	0
Soft Core (95–99%)	0	0
Shell (15–95%)	6,619	2,454
Cloud (0–15%)	59,504	34,918
Total	66,123	37,372

Table 5:

Top 10 Candidate Genes by F1 Score: Thermophile Cohort						
Gene	**Extremophile Ratio**	**Mesophile Ratio**	**Std. Residual**	**F1 Score**	**Prokka Annotation**	**Functional Category**
ctsr~~~ctsR	1	0.6000000	1.345	0.759	hypothetical protein;Transcriptional regulator CtsR	Other/Uncategorized
mcsB	1	0.6000000	1.345	0.759	Protein-arginine kinase	Metabolism
rpmJ	1	0.8666667	0.975	0.708	Large ribosomal subunit protein bL36	Translation/Ribosomal
rpsM	1	1.0000000	NaN	0.667	Small ribosomal subunit protein uS13	Translation/Ribosomal
adk	1	1.0000000	NaN	0.667	Adenylate kinase	Metabolism
secY~~~secY_1	1	1.0000000	NaN	0.667	Protein translocase subunit SecY	Transport
rplO	1	1.0000000	NaN	0.667	Large ribosomal subunit protein uL15	Translation/Ribosomal
rpmD	1	1.0000000	NaN	0.667	Large ribosomal subunit protein uL30	Translation/Ribosomal
rpsE	1	1.0000000	NaN	0.667	Small ribosomal subunit protein uS5	Translation/Ribosomal
rpsL	1	1.0000000	NaN	0.667	Small ribosomal subunit protein uS12	Translation/Ribosomal

Table 6:

Top 10 Candidate Genes by F1 Score: Acidophile Cohort

Gene	Acidophile Ratio	Non-Acidophile Ratio	Std. Residual	F1 Score	Prokka Annotation	Functional Category
group_5909	1	0	3.317	1.000	Peptidoglycan-N-acetylglucosamine deacetylase	Cell Wall/Membrane
salA_1~~~salA~~~salA_2	1	0	3.317	1.000	Iron-sulfur cluster carrier protein	Other/Uncategorized
group_5648	1	0	3.317	1.000	RNA-binding protein YbxF	Other/Uncategorized
group_3590	1	0	3.317	1.000	tRNA pseudouridine synthase A	Metabolism; Translation/Ribosomal
group_6169	1	0	3.317	1.000	Cyclic di-AMP receptor A	Other/Uncategorized
group_6150	1	0	3.317	1.000	hypothetical protein	Other/Uncategorized
group_6149	1	0	3.317	1.000	Replication initiation control protein YabA	Other/Uncategorized
abrB	1	0	3.317	1.000	Transition state regulatory protein AbrB	Other/Uncategorized
group_6126	1	0	3.317	1.000	Methionine--tRNA ligase;hypothetical protein	DNA/RNA Processing; Translation/Ribosomal
group_6188	1	0	3.317	1.000	NAD-dependent malic enzyme	Other/Uncategorized

Figure 1:

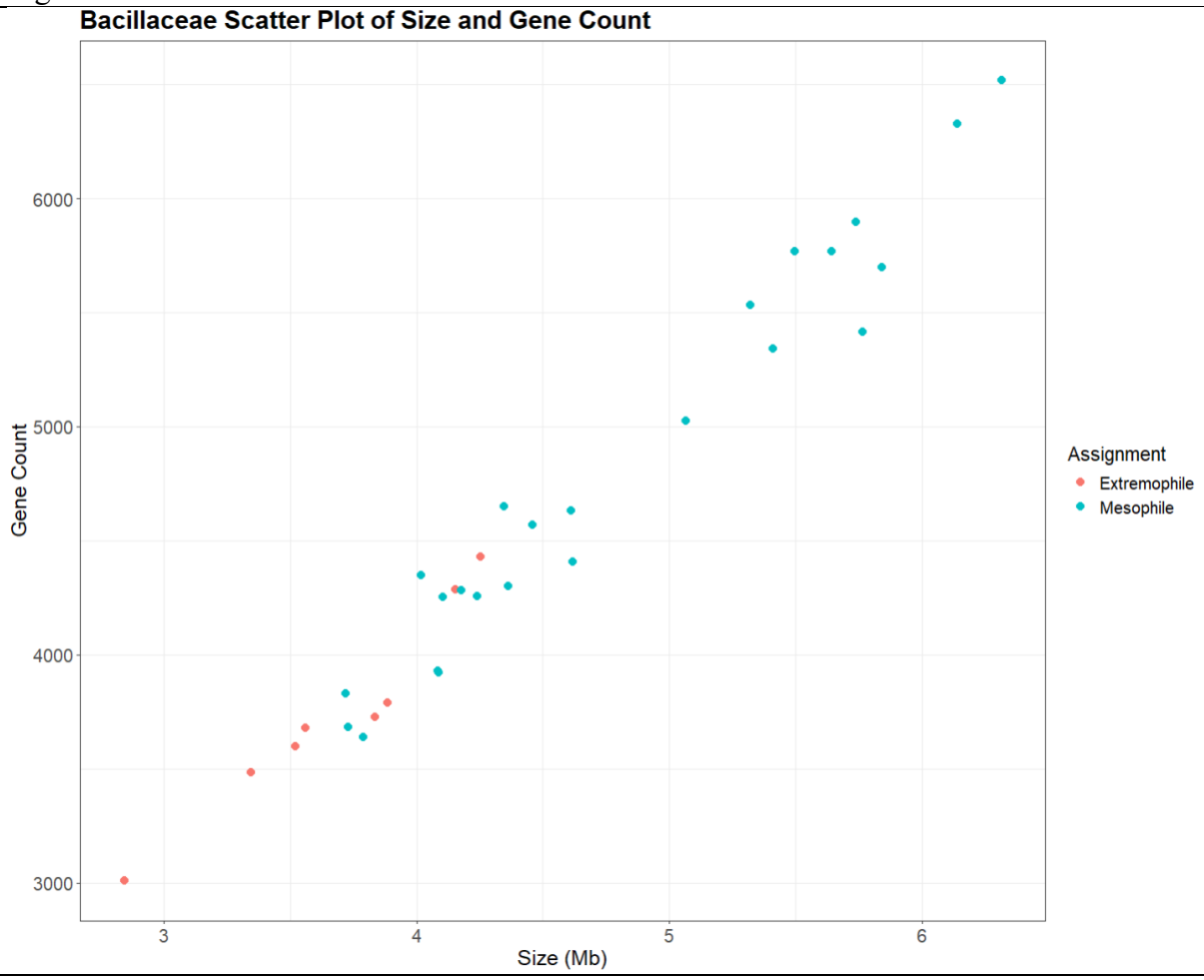

Bacillaceae Scatter Plot of Size and Gene Count
Gene Count
6000
5000
4000
3000
3
4
5
6
Size (Mb)
Assignment
Extremophile
Mesophile

Figure 2:

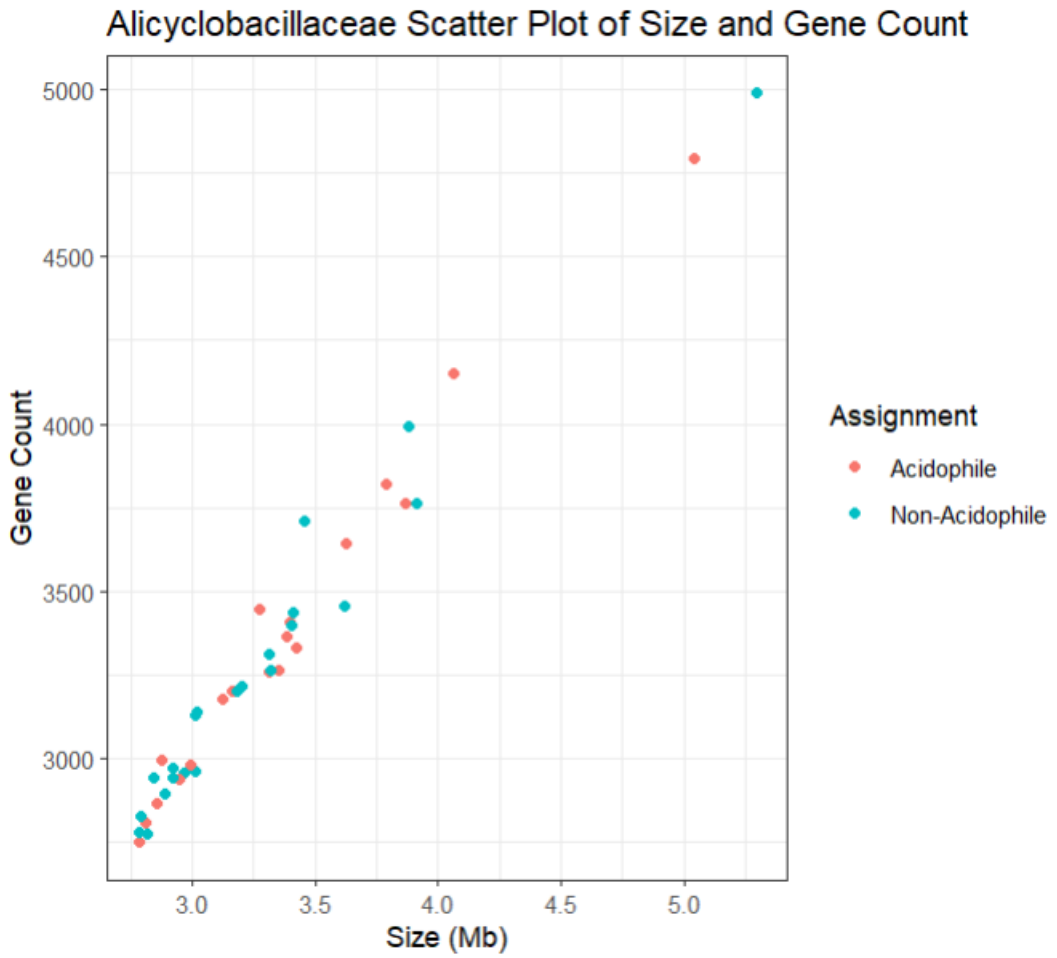

Alicyclobacillaceae Scatter Plot of Size and Gene Count
Gene Count
5000
4500
4000
3500
3000
3.0
3.5
4.0
4.5
5.0
Size (Mb)
Assignment
Acidophile
Non-Acidophile

Figure 3: Thermophile cohort random forest heatmap from *aurora*

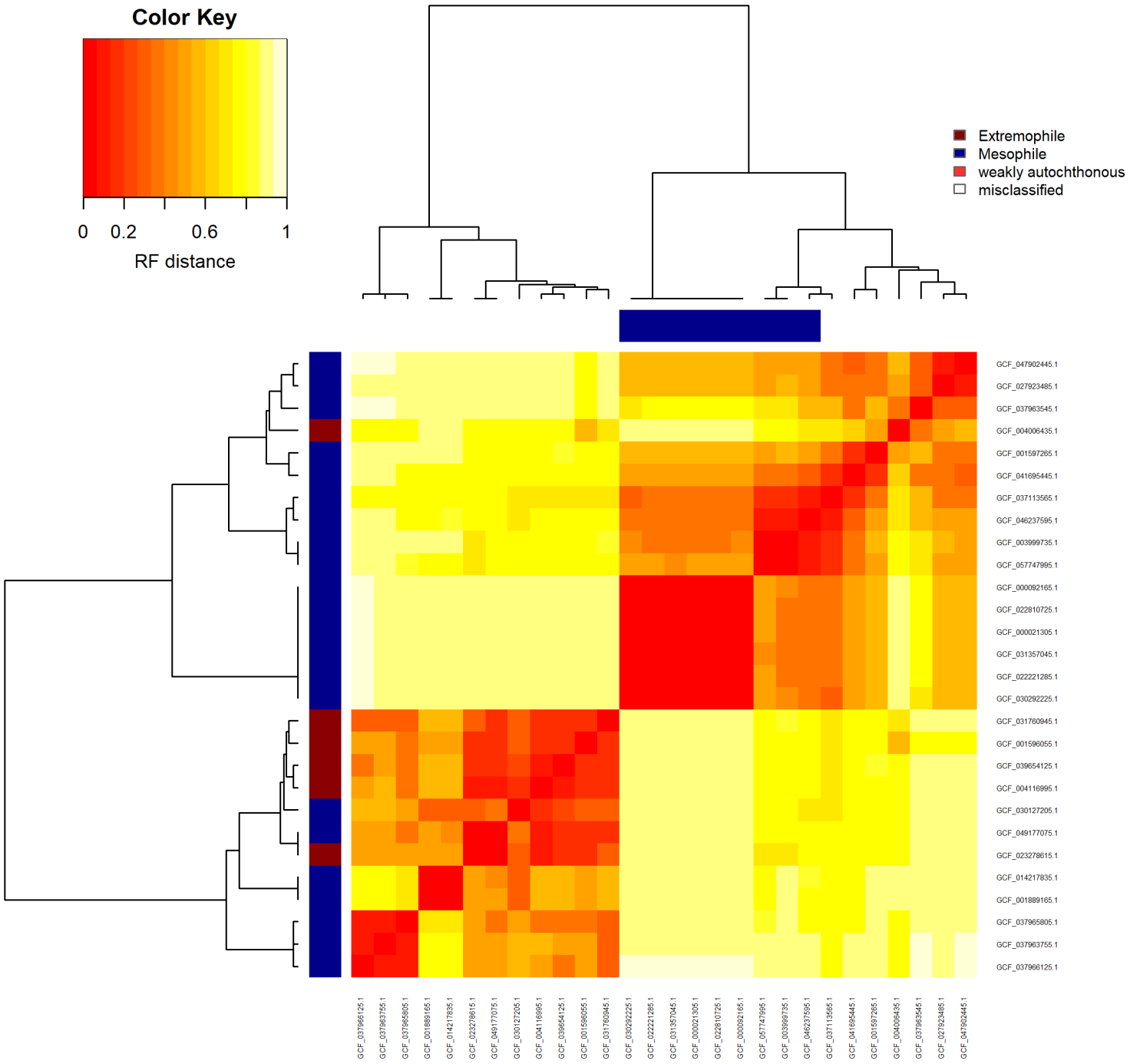
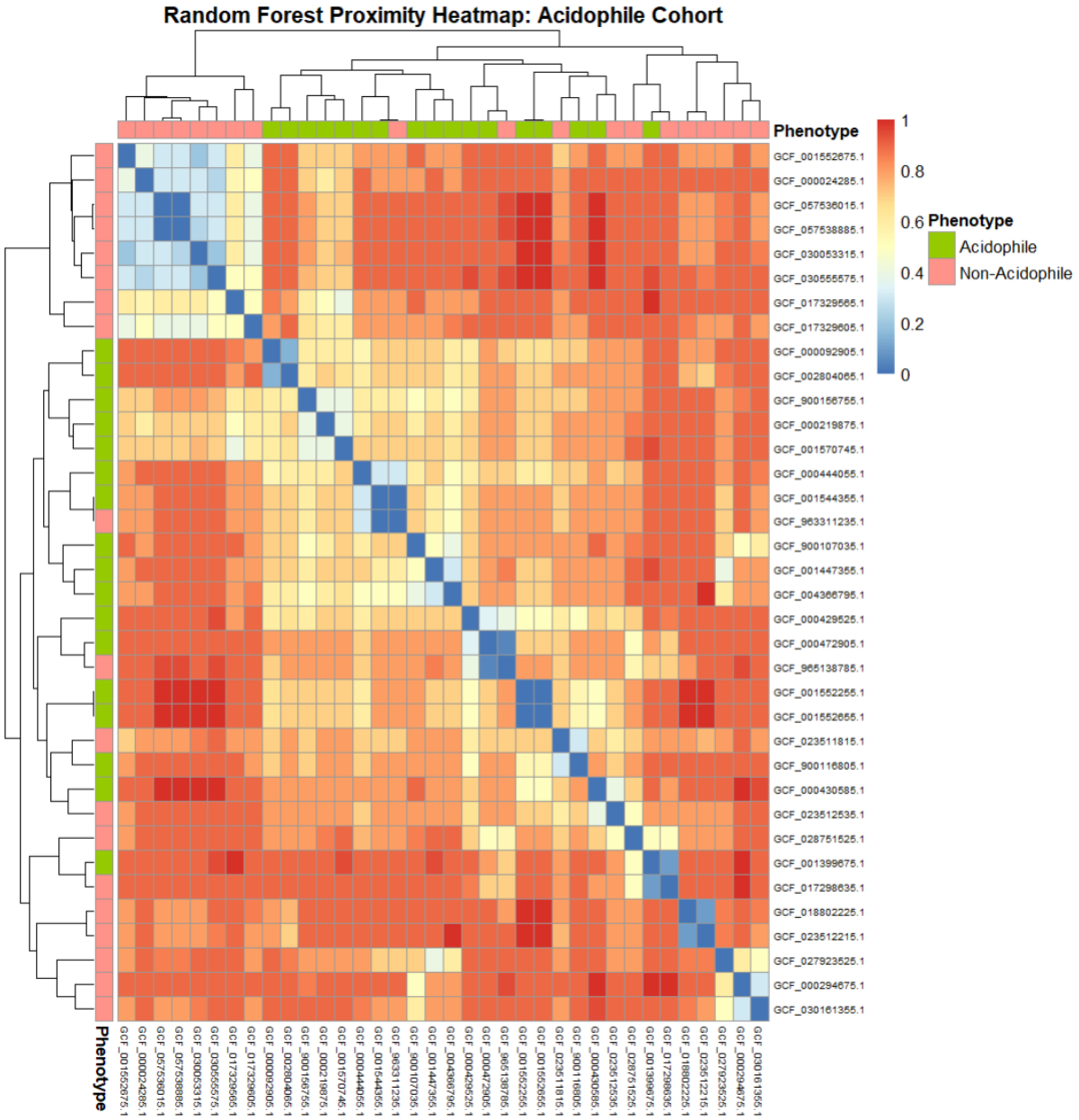


Figure 4:


Random Forest Proximity Heatmap: Acidophile Cohort
Phenotype
Acidophile
Non-Acidophile
1
0.8
0.6
0.4
0.2
0
GCF_001552675.1
GCF_000024285.1
GCF_057536015.1
GCF_057538885.1
GCF_030053315.1
GCF_030555575.1
GCF_017329565.1
GCF_017329605.1
GCF_000092905.1
GCF_002804065.1
GCF_900156755.1
GCF_000219875.1
GCF_001570745.1
GCF_000444055.1
GCF_001544355.1
GCF_963311235.1
GCF_900107035.1
GCF_001447355.1
GCF_004366795.1
GCF_000429525.1
GCF_000472905.1
GCF_965138785.1
GCF_001552255.1
GCF_001552655.1
GCF_023511815.1
GCF_900116805.1
GCF_000430585.1
GCF_023512535.1
GCF_028751525.1
GCF_001399675.1
GCF_017298635.1
GCF_018802225.1
GCF_023512215.1
GCF_027923525.1
GCF_000294675.1
GCF_030161355.1